

Past Paper 2024 Reconduct.

- ① Prove that for a hermitian operator the eigen vectors corresponding to different eigen value are orthogonal,
We know that as the given eigen function is

$$\hat{A}\psi_1 = a_1\psi_1 \quad \text{--- (1)}$$

$$\hat{A}\psi_2 = a_2\psi_2 \quad \text{--- (2)}$$

taking conjugate - (1) & (2)

$$A^*\psi_1^* = a_1^*\psi_1^* \quad \text{--- (3)}$$

$$A^*\psi_2^* = a_2^*\psi_2^* \quad \text{--- (4)}$$

As the hermitian operator is

$$\int_{-\infty}^{\infty} \psi_1^* \hat{A} \psi_2 dt = \int_{-\infty}^{\infty} \hat{A} \psi_1^* \psi_2 dt$$

$$\int_{-\infty}^{\infty} \psi_1^* a_2 \psi_2 dt = \int_{-\infty}^{\infty} a_1 \psi_1^* \psi_2 dt$$

$$a_2 \int_{-\infty}^{\infty} \psi_1^* \psi_2 dt = a_1 \int_{-\infty}^{\infty} \psi_1^* \psi_2 dt$$

$$a_2 \int_{-\infty}^{\infty} \psi_1^* \psi_2 dt - a_1 \int_{-\infty}^{\infty} \psi_1^* \psi_2 dt = 0$$

$$(a_2 - a_1) \int_{-\infty}^{\infty} \psi_1^* \psi_2 dt = 0$$

$$a_2 - a_1 \neq 0$$

So $\int_{-\infty}^{\infty} \psi_1^* \psi_2 dt = 0$
function is orthogonal $\rightarrow$

- ② What is Correspondence Principle.

The Correspondence principle is state that the Quantum number must be agree in classical theory in the limit of large Quantum number are known as correspondence principle.

$$E_n = 13.6 eV$$

$$n^2$$

③ Prove that $[L_x, L^2] = 0$

We know that

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

So

$$\begin{aligned} [L_x, L^2] &= [L_x, L_x^2 + L_y^2 + L_z^2] \\ [L_x, L^2] &= [L_x, L_x^2] + [L_x, L_y^2] + [L_x, L_z^2] \\ [L_x, L^2] &= [L_x, L_x L_x] + [L_x, L_y L_y] + [L_x, L_z L_z] \\ [L_x, L^2] &= \cancel{[L_x, L_x] L_x} + \cancel{[L_x, L_x] L_x} + [L_x, L_y] L_y + [L_x, L_y] L_y \\ &\quad + [L_x, L_z] L_z + [L_x, L_z] L_z \end{aligned}$$

$$\begin{aligned} [L_x, L^2] &= [L_x, L_y] L_y + [L_x, L_y] L_y + [L_x, L_z] L_z + [L_x, L_z] L_z \\ // &= \cancel{i\hbar L_z L_y} + \cancel{i\hbar L_x L_y} - \cancel{i\hbar L_y L_z} - \cancel{i\hbar L_y L_z} \end{aligned}$$

$$[L_x, L^2] = 0 \text{ Hence prove.}$$

④ Difference b/w bounded system and unbounded system.

Bounded system

When a particle are move in system they can't across to system on potential wall because the particle has less energy due to potential wall.

That's why they can't away for the system
→ It spectrum will be discrete.

Un bounded system.

When a particle are cross to the system on ~~due to~~ potential wall due to high energy of particle this system are collect unbounded system.

It spectrum will be continuous.

⑥ Define unitary operator:

A linear operator $\hat{U}$ is said to be unitary if its

inverse $\hat{U}^{-1}$ is equal to its adjoint
ie $\hat{U}^\dagger$

$$\hat{U}^\dagger = \hat{U}^{-1} \quad \text{or} \quad \hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = I$$

- ⑥ Write any two postulates of Quantum mechanics?
- Quantum system is completely described by a wave function which contains all information about system. It is denoted by ψ
 - Every measurable physical quantity is represented by a hermitian operator like as position, momentum, energy

① Prove that

Expectation value of hermitian operator is real

Define

$$\int \psi^* \hat{A} \psi dx = \int \hat{A}^\dagger \psi^* \psi dx \quad \leftarrow \text{Hermitian operator}$$

$$\langle \hat{A} \rangle = \int \psi^* \hat{A} \psi dx \quad \leftarrow \text{expectation value}$$

Solve

$$\langle \hat{A} \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{A} \psi(x) dx$$

taking complex

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} (\psi^* \hat{A} \psi)^* dx$$

$$\langle \hat{A} \rangle^* = \int_{-\infty}^{\infty} (\hat{A} \psi)^* \psi dx$$

use hermitian operator is real

$$\int_{-\infty}^{\infty} (\hat{A} \psi)^* \psi dx = \int \psi^* \hat{A} \psi dx$$

so

$$\langle \hat{A} \rangle^* = \langle \hat{A} \rangle$$

Here prove the expectation value

of hermitian operator is real.

→ Expectation value of anti hermitian operator is imaginary.

Solved

We know that expectation value is given that

$$\langle \hat{A} \rangle = \int \psi^* (\hat{A}) \psi dx \quad \text{--- (1) Conjugate}$$

$$\langle \hat{A} \rangle^* = \int (\psi^*)^* (\hat{A})^* \psi dx \quad \text{--- (2)}$$

$$\langle \hat{A} \rangle^* = \int (\hat{A} \psi)^* \psi dx \quad \text{Anticommutation operators}$$

$\hat{A}^* = -\hat{A}$

So

$$\int (\hat{A} \psi)^* \psi dx = \int \psi^* (-\hat{A}) \psi dx$$

$$\int (\hat{A} \psi)^* \psi dx = - \int \psi^* \hat{A} \psi dx$$

$$\langle \hat{A} \rangle^* = - \langle \hat{A} \rangle \quad \text{here proved}$$

⑧ Write an expression for identity operator and completeness relation for Dirac notation.

Dirac notation.

Dirac notation is a Quantum system to be represented by a Hilbert Space. So the Quantum state like as ket ($|\psi\rangle$) and bra ($\langle\psi|$) symbols.

Identity operator.

An operator which not change to original state are known identity operator.

Completeness Relation

$$\sum_n |n\rangle \langle n| = \hat{I}$$

⑨ Can we determine x-component of angular momentum and azimuthal angle with precision simultaneously?

انہما جہاں پر azimuthal آجائے تو اس کو ہم ϕ سے ظاہر کرتے ہیں۔

$\phi \rightarrow$ azimuthal.
 $L_z \rightarrow$ angle component

$$[L_z, \phi]$$

we know that

$$[L_z, \phi] \psi = [L_z \phi - \phi L_z] \psi$$

$$= \left[-i\hbar \frac{\partial}{\partial \phi} \phi \psi - \phi (-i\hbar) \frac{\partial \psi}{\partial \phi} \right]$$

$$= \left[-i\hbar \psi - i\hbar \phi \frac{\partial \psi}{\partial \phi} + i\hbar \phi \frac{\partial \psi}{\partial \phi} \right] = 0$$

Apply differential.

سوال میں جو چیزوں کے بارے میں پوچھا گیا ہے تو اس میں ہم اثر ϕ کو L_z پر کیا اثر ہے اس سے

پیمائش کریں تو ہم پیمائش کرنا چاہتے ہیں
 نیز اس کے اور اثر ہم ϕ کو پیمائش کرنا چاہتے ہیں
 نیز یہ کیونکہ $[L_z, \phi] = -i\hbar \phi$ Hence proved

Uncertainty Principle کے خلاف ہے تو ہم دونوں چیزوں کو ایک ساتھ پیمائش نہیں کر سکتے۔
 $\Delta \phi \Delta L_z \geq \hbar/2$

چونکہ ان دونوں کے درمیان nonzero ہے۔

10) Prove the following properties of Dirac Notation

i) Schwarz inequality ii) Triangle inequality

In any two Quantum state $|\psi\rangle$ and $|\phi\rangle$ of Hilbert space

We can show that in

Dirac Notation is

$$|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \cdot \langle \psi | \psi \rangle$$

and eigen function

$$|\psi\rangle = \alpha |\phi\rangle$$

α is a scalar quantity

$\rightarrow$ This inequality has

a linearly independent

vector Q.S. دو مختلف

مقدار inner product کی

product سے (جو) inner product

cosine, angle کی limit

کی طرف ہے

Triangle inequality gives a bound on the norm of a sum of vectors.

In Dirac Notation

$$|\psi\rangle + |\phi\rangle \leq |\psi\rangle + |\phi\rangle$$

$$|\psi + \phi|^2 \leq (|\psi|^2 + |\phi|^2)$$

یہ دو vectors کے جو sum کرنے کے بعد جو قابل آتی ہے

وہ ہمیشہ ان دونوں کی

مجموع سے

کم یا برابر ہوتی ہے۔

(11) What is the condition for which the sum of two projection operators $\hat{P}_1$ and $\hat{P}_2$ is also a projection operator.

In given conditions when the two projection operators are sum each other but they cannot return as original condition, so they are not a projection operator.

(12) Statement of principle of superposition -

When two or more wave moving through the same region of space, waves will superimpose and produce a well defined combined effect.

If $\psi = a_1 \psi_1 + a_2 \psi_2$ is a solution here $a_1, a_2 = \text{const}$. Then ψ_1 & ψ_2 follow the superposition principle. That means the net amplitude caused by two or more wave travelling same space is sum of amplitudes that would have been produced by individual wave separately.

$$\hat{O}\psi = \lambda\psi$$

(13) Write down the expression for the momentum operator in the position representation.

The momentum operator in the position representation is given by

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}$$

It acts on the wave function $\psi(x)$ as

$$\hat{p}\psi(x) = -i\hbar \frac{\partial}{\partial x} \psi(x)$$

(14) Write down the equation of continuity showing the conservation of probability in Q.M.

آئیے یاد رکھتے ہیں کہ (س) کا جواب مندرجہ ذیل ہے۔
آئیے یاد رکھیں کہ (س) کو ثابت کرنے کے لیے Schrodinger
time depend eq. استعمال کرتے ہیں۔

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = i\hbar \frac{\partial \psi}{\partial t} \quad \text{--- (1)} \quad \text{complex conjugate}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi^* + V\psi^* = -i\hbar \frac{\partial \psi^*}{\partial t} \quad \text{--- (2)}$$

xy by eq 1 in ψ^* & eq 2 in ψ with both are subtract

$$\frac{-\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = i\hbar \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right)$$

~~xy~~ $\div \hbar$ on b.s.

$$\frac{-\hbar}{2im} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = i \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right)$$

Divided by i on b.s.

$$\frac{-\hbar}{2im} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right)$$

$$\frac{-\hbar}{2im} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = \frac{\partial}{\partial t} (\psi^* \psi)$$

$$\frac{\partial}{\partial t} \psi^* \psi + \frac{\hbar}{2im} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) = 0$$

Compare by analyze

$$\frac{\partial \rho(\vec{r}, t)}{\partial t} + \nabla \cdot \vec{j} = 0$$

$\rho(\vec{r}, t) \rightarrow$ probability density
 $\vec{j} \rightarrow$ current density.

(15) Write down the expression for time rate of change of expectation value of an operator A .

The time rate of change of expectation value of an operator A is given by

$$\frac{d}{dt} \langle \hat{A} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle$$

اگر $C.M$ کی صورت میں دیکھا جائے تو اس condition میں
 ہم $\frac{d}{dt} \langle A \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle$ اور $\langle A \rangle$ کا فکشن یوتا
 سے تاریخ کا یٹن (جس میں اس میں فکشن) کرتے ہیں تو اس وقت
 اس کو $\frac{d}{dt} \langle A \rangle$ ضرب دیتے ہیں۔

$$\frac{d}{dt} \langle A \rangle = \frac{d}{dt} \int \psi^*(x,t) A \psi(x,t) dx$$

اور یٹن جن میں $\frac{d}{dt}$ کی صورت میں۔
 میری ہم $Schrodinger$ فکشن اس کو حل کرتے ہیں اور جواب

$$\frac{d}{dt} \langle A \rangle = \frac{i}{\hbar} \int \psi^* [\hat{H} \hat{A} - \hat{A} \hat{H}] \psi dx + \int \psi^* \frac{\partial A}{\partial t} \psi dx$$

$$\frac{d}{dt} \langle A \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle + \left(\frac{\partial A}{\partial t} \right)$$

Long Question

PUACP

$$P.E = \frac{qV}{4\pi\epsilon_0 r} = \frac{q^2}{4\pi\epsilon_0 r}$$

Central Potential:

A central Potential is a type of P.E that depends only on the distance from a central point (usually the origin).

$$V(\vec{r}) = V(r)$$

Solution of Radial eq -

We know that radial eq as given that (11)

$$\left[\frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dR}{dr} \right] + \left[\frac{2m}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) - \frac{l(l+1)}{r^2} \right] R = 0$$

$$\left[\frac{1}{r^2} \frac{d}{dr} \left[r^2 \frac{dR}{dr} \right] + \left[\frac{2mE}{\hbar^2} + \frac{2me^2}{4\pi\epsilon_0 \hbar^2 r} - \frac{l(l+1)}{r^2} \right] R = 0$$

$$\frac{1}{r^2} \left[r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} \right] + \left[\frac{2mE}{\hbar^2} + \frac{2}{r} \left(\frac{me^2}{4\pi\epsilon_0 \hbar^2} \right) - \frac{l(l+1)}{r^2} \right] R = 0$$

$$\frac{d^2 R}{dr^2} + 2 \frac{dR}{dr} + \left[\frac{2mE}{\hbar^2} + \frac{2}{r} \left(\frac{me^2}{4\pi\epsilon_0 \hbar^2} \right) - \frac{l(l+1)}{r^2} \right] R = 0$$

let (12)

$$-\alpha^2 = \frac{2mE}{\hbar^2} - 24(a)$$

$$h = \frac{me^2}{4\pi\epsilon_0 \hbar^2} \quad (24b)$$

$$h\alpha = \frac{me^2}{4\pi\epsilon_0 \hbar^2} \quad (24c)$$

24a, 24b, 24c put in 24

$$\frac{d^2 R}{dr^2} + 2 \frac{dR}{dr} + \left[-\alpha^2 + \frac{2h\alpha}{r} - \frac{l(l+1)}{r^2} \right] R = 0 \quad (25)$$

let

$$\rho = 2\alpha r$$

$$\left[\frac{dR}{d\rho} = 2\alpha \right]$$

Apply deriv -

$$\left[\frac{d}{dr} = \frac{d}{d\rho} \frac{d\rho}{dr} \right] \quad (26a)$$

$$\frac{dR}{dr} = \frac{dR}{d\rho} \frac{d\rho}{dr} \Rightarrow \left[\frac{dR}{dr} = \frac{dR}{d\rho} 2\alpha \right] \quad (26b)$$

again derivative

$$\frac{d^2 R}{dr^2} = \frac{d}{dr} \left(\frac{dR}{dr} \right) = \frac{d}{d\rho} \left(\frac{dR}{d\rho} 2\alpha \right) \cdot \frac{d\rho}{dr}$$

$$\left[\frac{d^2 R}{dr^2} = 4\alpha \frac{d^2 R}{d\rho^2} \right] \quad (26c)$$

Putty 26a, 26b, 26c in 25

$$4\alpha^2 \frac{d^2 R}{ds^2} + \frac{4\alpha}{s} \left(2\alpha \frac{dR}{ds} \right) + \left[-\alpha^2 + \frac{4\alpha^2}{s} - \frac{\ell(\ell+1)}{(s/2\alpha)^2} \right] R = 0$$

$$4\alpha^2 \frac{d^2 R}{ds^2} + \frac{8\alpha^2}{s} \frac{dR}{ds} + \left[-\alpha^2 + \frac{4\alpha^2}{s} - \frac{4\alpha^2 \ell(\ell+1)}{s^2} \right] R = 0$$

$$\frac{d^2 R}{ds^2} + \frac{2}{s} \frac{dR}{ds} + \left[\frac{-1}{4} + \frac{1}{s} - \frac{1}{s^2} \ell(\ell+1) \right] R = 0 \quad (27)$$

اگر $s \rightarrow \infty$ تو $\frac{1}{s} \rightarrow 0$ تو $\frac{1}{s^2} \rightarrow 0$ تو $\frac{1}{s^2} \ell(\ell+1) \rightarrow 0$ تو $\frac{1}{4}$ باقی می ماند

$$\boxed{\frac{d^2 R}{ds^2} - \frac{R}{4} = 0} \quad (28)$$

$R(s) = N e^{s/2}$
form.

Similarly apply
same eq.
when N is Normalization

The exact solution is given by

$$\rightarrow \frac{d^2 R}{ds^2} + \frac{2}{s} \frac{dR}{ds} + \left[\frac{-1}{4} + \frac{1}{s} - \frac{\ell(\ell+1)}{s^2} \right] R = 0$$

We know that R is function of s and $F(u)$

$$\rightarrow R(s) = F(u) N e^{-s/2}$$

$$R = N e^{-s/2} \cdot f \quad (29a)$$

Apply derivative

$$\frac{dR}{ds} = N \left[e^{-s/2} \frac{df}{ds} + e^{-s/2} \left(-\frac{1}{2} \right) f \right]$$

$$\frac{dR}{ds} = N \left[e^{-s/2} \frac{df}{ds} - \frac{e^{-s/2}}{2} f \right] \quad (29b)$$

again we

$$\frac{d^2 R}{ds^2} = N \left[e^{-s/2} \frac{d^2 f}{ds^2} + e^{-s/2} \left(-\frac{1}{2} \right) \frac{df}{ds} - \frac{1}{2} \left(e^{-s/2} \frac{df}{ds} + e^{-s/2} \left(-\frac{1}{2} \right) f \right) \right]$$

$$\frac{d^2 R}{ds^2} = N \left[e^{-s/2} \frac{d^2 f}{ds^2} - \frac{1}{2} e^{-s/2} \frac{df}{ds} - \frac{1}{2} e^{-s/2} \frac{df}{ds} + \frac{1}{4} e^{-s/2} f \right]$$

$$\frac{d^2 R}{ds^2} = N \left[e^{-s/2} \frac{d^2 f}{ds^2} - e^{-s/2} \frac{df}{ds} + \frac{1}{4} e^{-s/2} f \right] \quad (29c)$$

Putty value 29a, 29b, 29c in 27

$$N \left[e^{-s/2} \frac{d^2 F}{ds^2} - e^{-s/2} \frac{dF}{ds} + \frac{1}{4} e^{s/2} F \right] + 2N \left[e^{-s/2} \frac{dF}{ds} - \frac{1}{2} e^{-s/2} F \right] + \left[\frac{1}{4} + \frac{1}{s} - \frac{l(l+1)}{s^2} \right] N e^{-s/2} F = 0$$

Divide by $N e^{-s/2}$

$$\frac{d^2 F}{ds^2} - \frac{dF}{ds} + \frac{1}{4} F + \frac{2}{s} \left[\frac{dF}{ds} - \frac{1}{2} F \right] + \left[\frac{1}{4} + \frac{1}{s} - \frac{l(l+1)}{s^2} \right] F = 0$$

$$\frac{d^2 F}{ds^2} - \frac{dF}{ds} + \frac{1}{4} F + \frac{2}{s} \frac{dF}{ds} - \frac{F}{s} - \frac{F}{4} + \frac{F}{s} - \frac{l(l+1)F}{s^2} = 0$$

$$\frac{d^2 F}{ds^2} + \left(\frac{2-1}{s} \right) \frac{dF}{ds} + \left[\frac{1}{s} - \frac{1}{s} - \frac{l(l+1)}{s^2} \right] F = 0$$

find value

$$\left(\frac{d^2 F}{ds^2} \right) + \left(\frac{2-1}{s} \right) \left(\frac{dF}{ds} \right) + \left[\frac{1}{s} - \frac{1}{s} - \frac{l(l+1)}{s^2} \right] F = 0 \quad (30)$$

let

$$F(s) = s^l G(s) \quad (31a)$$

$$\frac{dF}{ds} = \left(s^l \frac{dG}{ds} + l s^{l-1} G \right) \quad (31b) \quad \text{Apply v.v rule}$$

$$\frac{d^2 F}{ds^2} = \left(s^l \frac{d^2 G}{ds^2} + l s^{l-1} \frac{dG}{ds} \right) + \left(l s^{l-1} \frac{dG}{ds} + l(l-1) s^{l-2} G \right)$$

$$\frac{d^2 F}{ds^2} = s^l \frac{d^2 G}{ds^2} + l s^{l-1} \frac{dG}{ds} + l s^{l-1} \frac{dG}{ds} + l(l-1) s^{l-2} G = 0 \quad (31c)$$

$$\left[s^l \frac{d^2 G}{ds^2} + 2l s^{l-1} \frac{dG}{ds} + l(l-1) s^{l-2} G \right] + \left(\frac{2-1}{s} \right) \left[s^l \frac{dG}{ds} + l s^{l-1} G \right] + \left[\frac{1}{s} - \frac{1}{s} - \frac{l(l+1)}{s^2} \right] s^l G = 0$$

$$s^l \frac{d^2 G}{ds^2} + 2l s^{l-1} \frac{dG}{ds} + l(l-1) s^{l-2} G + \left(\frac{2-1}{s} \right) \left[s^l \frac{dG}{ds} + l s^{l-1} G \right] + \left[\frac{1}{s} - \frac{l(l+1)}{s^2} \right] s^l G = 0$$

$\div s^{l-1}$ on b.s.

$$\left[\int \frac{d^2 G}{ds^2} + 2L \frac{dG}{ds} + \frac{l(l-1)}{s} G \right] + \left(\frac{2-l}{s} \right) \left[\int \frac{dG}{ds} + lG \right] + \left[\frac{l-1}{s} - \frac{l(l+1)}{s^2} \right] sG = 0 \quad s \text{ common } \div$$

$$\left[\int \frac{d^2 G}{ds^2} + 2L \frac{dG}{ds} + \frac{l(l-1)}{s} G \right] + \left(\frac{2-l}{s} \right) \left[\int \frac{dG}{ds} + lG \right] + \left[\frac{l-1}{s} - \frac{l(l+1)}{s} \right] G = 0 \quad \frac{dG}{ds} \text{ comm}$$

$$\int \frac{d^2 G}{ds^2} + \frac{dG}{ds} \left[2L + \left(\frac{2-l}{s} \right) s \right] + G \left[\frac{l(l-1)}{s} + \left(\frac{2-l}{s} \right) l + l-1 - \frac{l(l+1)}{s} \right] = 0$$

$$\int \frac{d^2 G}{ds^2} + \frac{dG}{ds} [2L + 2 - s] + G \left[\frac{l^2}{s} - \frac{l}{s} + \frac{2l}{s} - \frac{l+l-1-l^2-l}{s} \right] = 0$$

$$\int \frac{d^2 G}{ds^2} + [2L + 1 - s] \frac{dG}{ds} + [(l-1-l) + 1] \frac{dG}{ds} + (l-1-l)G = 0$$

$$\int \frac{d^2 G}{ds^2} + [(2L+1) + 1 - s] \frac{dG}{ds} + (l-1-l)G = 0$$

let $l = n$

$$\int \frac{d^2 G}{ds^2} + [(2L+1) + 1 - s] \frac{dG}{ds} + (n-1-l)G = 0$$

let

$$2L+1 = p \quad n+l = n'$$

then

$$n-p = n+l-p = n+l-(2L+1)$$

$$n'-p = n+l-2L-1$$

$$\boxed{\int \frac{d^2 G}{ds^2} + [p+1-s] \frac{dG}{ds} + (n'-p)G = 0} \quad (32)$$

This eq shows that Laguerre's differential eq is given as

$$x \frac{d^2}{dx^2} L_n^m(x) + (m+1-x) \frac{d}{dx} L_n^m(x) + (n-m) L_n^m(x) = 0 \quad (33)$$

The solution of eqⁿ (32) & (33) give the expression for $\ln(x)$ are called associated Laguerre polynomials.
Thus the solution of eqⁿ (32) is given as

$$G(s) = L_n^m(x) \Rightarrow G(p) = L_n^p(s)$$

$$G(s) = L_{n+1}^{2L+1}(s) \quad (3)$$

using 34 in 31 a we get

$$F(y) = \int^L l_{n+1}^{2L+1}(y) \quad \text{--- (25)}$$

Put 55 in 29g

$$R(s) = N e^{-s/2} \int_0^{\infty} L_{n+1}(s) ds \quad \text{--- (36)}$$

ہمیں یاد ہے کہ جو فٹنس تھا وہ ہمیں (۱۹۸۰ء) کے روزِ
خود سے مل گیا تھا اس نے ہم

$$J = 2\alpha\pi - (a)$$

$$\alpha = \frac{me^2}{4\pi\epsilon_0\hbar^2} = \alpha = \frac{me^2}{4\pi\epsilon_0\hbar^2}$$

$$f = \frac{2\pi}{h} \left(\frac{m e^2}{4\pi \epsilon_0 h^2} \right)$$

$$J = \frac{2\pi}{n} \left(\frac{me^2}{4\pi\epsilon_0 \hbar^2} \right)$$

$$\therefore a = \frac{4\pi \epsilon_0 \hbar^2}{m e^2} \text{ is known as Bohr's orbit}$$

$$J = \frac{2\pi}{n} \frac{1}{a_1} \Rightarrow J = \frac{2\pi}{n a_1} \quad (37)$$

using 37 in 36 we get

$$R(n) = N e^{-S/2} \left(\frac{2\pi}{na} \right)^{1/2} e^{2\pi i} \left(\frac{2\pi}{na} \right)^{1/2}$$

$$R(n) = N e^{-\left(\frac{2n}{n_0}\right)^2} \left(\frac{2n}{n_0}\right)^{2L+1} \left(\frac{2n}{n_0}\right)$$

$$R(n) = Ne^{\frac{1}{2n}} \left(\frac{2n}{n-1} \right)^2 \left(\frac{2n}{n-1} \right)^{2n-1} \dots \quad (38)$$

Normalisation condition: $\int_0^\infty R^2(r) r^2 dr = 1$

کی رقم same ہوگی یعنی $R^n = R$

$$N = \frac{\left(\frac{2}{n_0}\right)^3 (n-l-1)!}{2n(n+l)!} \quad \text{--- (5)}$$

Putting 39 in 38

$$R(n) = \left(\frac{2}{n\alpha} \right)^3 \frac{(n-l+1)}{2n\{n+l\}^3} e^{-2/n\alpha} \left(\frac{2n}{n\alpha} \right)^l l^{2l+1} \left(\frac{2n}{n\alpha} \right)$$

This is called radial equation.

The necessary condition to obtain the above solution is that $n \geq l+1$
 n is principle quantum number.

$$n = 1, 2, 3, \dots, l+1$$

$$Q^2 = -\frac{2m\epsilon}{\hbar^2}$$

$$\left(\frac{me^2}{4\pi\epsilon_0\hbar^2}\right)^2 = -\frac{2mb}{b^2}$$

$$E = -\frac{h^2 m e^4}{2m (4\pi\epsilon_0)^2 h^2 n^2}$$

$$E_n = - \frac{13.6}{n^2} \text{ eV}$$

which is in accordance with Bohm's model of H-atom.

General

Uncertainty relation b/w two operators

There are two different hermite operators as given below

$$\hat{A} = (\hat{A}^\dagger)$$

$$\hat{B} = (\hat{B}^\dagger)$$

$$\langle \Delta A \rangle = \langle \hat{A} \rangle - \langle \hat{A} \rangle$$

$$\langle \Delta B \rangle = \langle \hat{B} \rangle - \langle \hat{B} \rangle$$

$$\langle (\Delta A)^2 \rangle = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2$$

$$\langle (\Delta B)^2 \rangle = \langle \hat{B}^2 \rangle - \langle \hat{B} \rangle^2$$

let

$$\langle \psi | \psi \rangle \geq 0$$

$$\langle \phi | (\Delta \hat{A} - i\lambda \Delta \hat{B})(\Delta \hat{A} + i\lambda \Delta \hat{B}) | \phi \rangle \geq 0$$

$$\langle \phi | (\Delta \hat{A})^2 + i\lambda \Delta \hat{A} \Delta \hat{B} - i\lambda \Delta \hat{B} \Delta \hat{A} - i^2 \lambda^2 (\Delta \hat{B})^2 | \phi \rangle \geq 0$$

$$\langle \phi | (\Delta \hat{A})^2 + i\lambda [\hat{A}, \hat{B}] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle \geq 0 \quad \text{--- (1)}$$

apply differential w.r.t

$$\frac{d}{d\lambda} \langle \phi | (\Delta \hat{A})^2 + i\lambda [\hat{A}, \hat{B}] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle = 0$$

$$\frac{d}{d\lambda} \langle \phi | i\lambda [\hat{A}, \hat{B}] + \lambda^2 (\Delta \hat{B})^2 | \phi \rangle = 0$$

$$\langle \phi | i[\hat{A}, \hat{B}] | \phi \rangle + 2\lambda (\Delta \hat{B})^2 = 0$$

$$\lambda = \frac{-i\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle}{2\langle (\Delta \hat{B})^2 \rangle}$$

putting value λ in (1)

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{2\langle (\Delta \hat{B})^2 \rangle} + \left(\frac{-i\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle}{2\langle (\Delta \hat{B})^2 \rangle} \right)^2$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{2\langle (\Delta \hat{B})^2 \rangle} - \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4\langle (\Delta \hat{B})^2 \rangle}$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{2\langle (\Delta \hat{B})^2 \rangle} - \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4\langle (\Delta \hat{B})^2 \rangle}$$

$$\langle (\Delta \hat{A})^2 \rangle + \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4\langle (\Delta \hat{B})^2 \rangle} \geq 0$$

$$\langle (\Delta \hat{A})^2 \rangle \geq \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4\langle (\Delta \hat{B})^2 \rangle}$$

$$\langle (\Delta \hat{A})^2 \rangle \langle (\Delta \hat{B})^2 \rangle \geq \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4}$$

$$\langle (\Delta \hat{A})^2 \rangle \langle (\Delta \hat{B})^2 \rangle \geq \frac{|\langle \phi | [\hat{A}, \hat{B}] | \phi \rangle|^2}{4}$$

$i^2 = -1$ $(-i^2) = (-1 \cdot i)^2$ $(-1)^2 (i^2) (1 \cdot 1) = -1$
--

$$\text{let } [\hat{A}\hat{B}] = i\hat{C}$$

$$\langle (\Delta\hat{A})^2 \rangle \langle (\Delta\hat{B})^2 \rangle = -\frac{|\langle \phi | i\hat{C} | \phi \rangle|^2}{4}$$

$$\langle (\Delta\hat{A})^2 \rangle \langle (\Delta\hat{B})^2 \rangle = -\frac{1}{4} |\langle \phi | \hat{C} | \phi \rangle|^2$$

$$\langle (\Delta\hat{A})^2 \rangle \langle (\Delta\hat{B})^2 \rangle = \frac{1}{4} |\langle \phi | \hat{C} | \phi \rangle|^2$$

Taking s.v.b.s

$$\langle \Delta\hat{A} \Delta\hat{B} \rangle = \frac{1}{2} \langle \hat{C} \rangle \quad \text{hence proved}$$

Find the eigen function and correspondence eigenvalue for a 1 dimensional harmonic oscillator using the algebraic method and prove the energy spectrum both discrete of degeneration

June 17, 2025

1 Introduction

This document mathematically proves that the energy spectrum of the 1-dimensional quantum harmonic oscillator (1D QHO) is both discrete and non-degenerate. We will utilize the algebraic method, employing ladder operators, which offers an elegant solution to this fundamental problem in quantum mechanics.

2 Hamiltonian of the Harmonic Oscillator

The Hamiltonian $\hat{H}$ for a 1D QHO is given by:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where $\hat{x}$ is the position operator, $\hat{p}$ is the momentum operator, m is the mass of the particle, and ω is the angular frequency of the oscillator.

The fundamental commutation relation between position and momentum operators is:

$$[\hat{x}, \hat{p}] = i\hbar$$

2.1 Definition of Ladder Operators

We define the annihilation operator ($\hat{a}$) and the creation operator ($\hat{a}^\dagger$) as:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega}\hat{p} \right)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega}\hat{p} \right)$$

These operators satisfy the commutation relation:

$$[\hat{a}, \hat{a}^\dagger] = 1$$

Let's prove this commutation relation:

$$\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= \left[\sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right), \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \right] \\
&= \frac{m\omega}{2\hbar} \left[\hat{x} + \frac{i}{m\omega} \hat{p}, \hat{x} - \frac{i}{m\omega} \hat{p} \right] \\
&= \frac{m\omega}{2\hbar} \left([\hat{x}, \hat{x}] - \left[\hat{x}, \frac{i}{m\omega} \hat{p} \right] + \left[\frac{i}{m\omega} \hat{p}, \hat{x} \right] - \left[\frac{i}{m\omega} \hat{p}, \frac{i}{m\omega} \hat{p} \right] \right) \\
&= \frac{m\omega}{2\hbar} \left(0 - \frac{i}{m\omega} [\hat{x}, \hat{p}] + \frac{i}{m\omega} [\hat{p}, \hat{x}] - 0 \right) \\
&= \frac{m\omega}{2\hbar} \left(-\frac{i}{m\omega} (i\hbar) + \frac{i}{m\omega} (-i\hbar) \right) \\
&= \frac{m\omega}{2\hbar} \left(\frac{\hbar}{m\omega} + \frac{\hbar}{m\omega} \right) \\
&= \frac{m\omega}{2\hbar} \left(\frac{2\hbar}{m\omega} \right) \\
&= 1
\end{aligned}$$

3 Hamiltonian in Terms of Ladder Operators

Let's express the Hamiltonian in terms of $\hat{a}$ and $\hat{a}^\dagger$. Consider the product $\hat{a}^\dagger \hat{a}$:

$$\begin{aligned}
\hat{a}^\dagger \hat{a} &= \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) \\
&= \frac{m\omega}{2\hbar} \left(\hat{x}^2 + \frac{i}{m\omega} \hat{x} \hat{p} - \frac{i}{m\omega} \hat{p} \hat{x} - \frac{i^2}{m^2 \omega^2} \hat{p}^2 \right) \\
&= \frac{m\omega}{2\hbar} \left(\hat{x}^2 + \frac{i}{m\omega} (\hat{x} \hat{p} - \hat{p} \hat{x}) + \frac{1}{m^2 \omega^2} \hat{p}^2 \right) \\
&= \frac{m\omega}{2\hbar} \left(\hat{x}^2 + \frac{i}{m\omega} [\hat{x}, \hat{p}] + \frac{1}{m^2 \omega^2} \hat{p}^2 \right)
\end{aligned}$$

Substitute $[\hat{x}, \hat{p}] = i\hbar$:

$$\begin{aligned}
\hat{a}^\dagger \hat{a} &= \frac{m\omega}{2\hbar} \left(\hat{x}^2 + \frac{i}{m\omega} (i\hbar) + \frac{1}{m^2 \omega^2} \hat{p}^2 \right) \\
&= \frac{m\omega}{2\hbar} \left(\hat{x}^2 - \frac{\hbar}{m\omega} + \frac{1}{m^2 \omega^2} \hat{p}^2 \right) \\
&= \frac{m\omega}{2\hbar} \hat{x}^2 - \frac{1}{2} + \frac{1}{2m\hbar\omega} \hat{p}^2
\end{aligned}$$

Now, rearrange this to connect with the Hamiltonian:

$$\begin{aligned}\hat{a}^\dagger \hat{a} + \frac{1}{2} &= \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{1}{2m\hbar\omega} \hat{p}^2 \\ \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) &= \hbar\omega \left(\frac{m\omega}{2\hbar} \hat{x}^2 + \frac{1}{2m\hbar\omega} \hat{p}^2 \right) \\ &= \frac{1}{2} m\omega^2 \hat{x}^2 + \frac{1}{2m} \hat{p}^2\end{aligned}$$

This is exactly the Hamiltonian $\hat{H}$. Thus, we can write:

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

We define the ****number operator**** $\hat{N}$ as:

$$\hat{N} = \hat{a}^\dagger \hat{a}$$

So the Hamiltonian becomes:

$$\hat{H} = \hbar\omega \left(\hat{N} + \frac{1}{2} \right)$$

4 Finding the Energy Spectrum (Eigenvalues)

Let's find the eigenstates of $\hat{N}$. Assume there exists an eigenstate $|n\rangle$ such that:

$$\hat{N} |n\rangle = n |n\rangle$$

where n is the eigenvalue. Applying the Hamiltonian to this eigenstate:

$$\hat{H} |n\rangle = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) |n\rangle$$

$$\hat{H} |n\rangle = \hbar\omega \left(n + \frac{1}{2} \right) |n\rangle$$

So the energy eigenvalues are:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

4.1 Proof of Discreteness

To show that n must be a non-negative integer, we use the properties of the ladder operators. First, consider the expectation value of the number operator:

$$\langle \hat{N} \rangle = \langle \psi | \hat{a}^\dagger \hat{a} | \psi \rangle = \langle \hat{a} | \psi \rangle \langle \hat{a} | \psi \rangle = \| \hat{a} | \psi \rangle \|^2 \geq 0$$

Since the norm squared of any state must be non-negative, the eigenvalues n of $\hat{N}$ must also be non-negative ($n \geq 0$).

Next, consider the action of the ladder operators on an eigenstate $|n\rangle$:

$$\begin{aligned}\hat{N}(\hat{a}|n\rangle) &= \hat{a}^\dagger \hat{a} \hat{a}|n\rangle \\ &= (\hat{a} \hat{a}^\dagger - 1) \hat{a}|n\rangle \quad (\text{since } \hat{a}^\dagger \hat{a} = \hat{a} \hat{a}^\dagger - 1) \\ &= \hat{a}(\hat{a}^\dagger \hat{a} - 1)|n\rangle \\ &= \hat{a}(\hat{N} - 1)|n\rangle \\ &= \hat{a}(n - 1)|n\rangle \\ &= (n - 1)(\hat{a}|n\rangle)\end{aligned}$$

This shows that if $|n\rangle$ is an eigenstate of $\hat{N}$ with eigenvalue n , then $\hat{a}|n\rangle$ is an eigenstate with eigenvalue $n - 1$. The operator $\hat{a}$ is thus an "annihilation" or "lowering" operator.

Similarly for $\hat{a}^\dagger$:

$$\begin{aligned}\hat{N}(\hat{a}^\dagger|n\rangle) &= \hat{a}^\dagger \hat{a} \hat{a}^\dagger|n\rangle \\ &= \hat{a}^\dagger(\hat{a}^\dagger \hat{a} + 1)|n\rangle \quad (\text{since } \hat{a} \hat{a}^\dagger = \hat{a}^\dagger \hat{a} + 1) \\ &= \hat{a}^\dagger(\hat{N} + 1)|n\rangle \\ &= \hat{a}^\dagger(n + 1)|n\rangle \\ &= (n + 1)(\hat{a}^\dagger|n\rangle)\end{aligned}$$

This shows that $\hat{a}^\dagger|n\rangle$ is an eigenstate with eigenvalue $n + 1$. The operator $\hat{a}^\dagger$ is a "creation" or "raising" operator.

Since $n \geq 0$, there must be a lowest possible eigenvalue. Let's assume this lowest state is $|0\rangle$, for which $\hat{a}$ acting on it yields zero (it cannot go lower):

$$\hat{a}|0\rangle = 0$$

Applying the number operator to this state:

$$\hat{N}|0\rangle = \hat{a}^\dagger \hat{a}|0\rangle = \hat{a}^\dagger(0) = 0$$

So, the lowest possible integer value for n is 0. By repeatedly applying $\hat{a}^\dagger$ to the ground state $|0\rangle$, we can generate all higher energy states:

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle$$

This confirms that n must take only non-negative integer values: $n = 0, 1, 2, 3, \dots$. Therefore, the energy values $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$ are discrete. The spacing between consecutive energy levels is constant:

$$E_{n+1} - E_n = \hbar\omega\left(\left(n + 1\right) + \frac{1}{2}\right) - \hbar\omega\left(n + \frac{1}{2}\right) = \hbar\omega$$

5 Proving Non-Degeneracy

Degeneracy means that two or more distinct quantum states have the same energy eigenvalue. In the 1D QHO, each energy level E_n is uniquely determined by the quantum number n . The ladder operator method uniquely constructs each eigenstate $|n\rangle$ from the ground state $|0\rangle$ by applying the creation operator $\hat{a}^\dagger$ n times. For a given value of n , there is only one corresponding state $|n\rangle$. There are no other independent quantum numbers in the 1D system that could lead to different states having the same energy. For example, in 3D systems, angular momentum quantum numbers can introduce degeneracy. However, in 1D, no such additional quantum numbers exist. Since each energy level E_n corresponds to only one unique quantum state $|n\rangle$, there is no degeneracy in the 1D quantum harmonic oscillator.

6 Conclusion

The energy spectrum of the 1D quantum harmonic oscillator is mathematically proven to be:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, 3, \dots$$

- **Discrete:** The quantum number n takes only non-negative integer values, leading to quantized energy levels with a constant spacing of $\hbar\omega$.
- **Non-degenerate:** Each energy level E_n corresponds to a unique quantum state $|n\rangle$, as there are no additional quantum numbers in one dimension to cause multiple states to have the same energy.

Wave functions of harmonic oscillator are shown in fig 3.3.

Wave functions: If we apply lowering operator $\hat{a}$ to lowest possible state ψ_0 , we get zero i.e.

in harmonic oscillator wave function is equal to eigen function so can be called as eigen function for SHO

$$\hat{a}\psi_0 = 0 \Rightarrow \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})\psi_0 = 0 \Rightarrow m\omega\hat{x}\psi_0 + i\left(-i\hbar\frac{d}{dx}\right)\psi_0 = 0$$

$$\Rightarrow m\omega x\psi_0 = -\hbar\frac{d\psi_0}{dx} \Rightarrow \frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar}x dx \Rightarrow \ln\psi_0 = -\frac{m\omega x^2}{2\hbar} + \ln A_0$$

$$\Rightarrow \ln\psi_0 = \ln e^{-\frac{m\omega x^2}{2\hbar}} + \ln A_0 \Rightarrow \ln\psi_0 = \ln A_0 e^{-\frac{m\omega x^2}{2\hbar}} \Rightarrow \psi_0 = A_0 e^{-\frac{m\omega x^2}{2\hbar}}$$

Put $\frac{m\omega}{\hbar} = \alpha$, then

$$\psi_0 = A_0 e^{-\alpha x^2/2}$$

By applying normalization condition, we can find A_0 . So normalized wave function for ground state of harmonic oscillator is,

$$\psi_0 = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

This is a brief theory of harmonic oscillator.